

PVsyst - Simulation report

Grid-Connected System

Project: DHL

Variant: C&I rooftop – Munich (on DHL Paketzentrum Aschheim)

Sheds on ground

System power: 687 kWp

LessonNo 1 - Germany



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PVsyst V8.1.4

VCO, Simulation date:
07/07/26 00:24
with V8.1.4

Project summary

Geographical Site

LessonNo 1

Germany

Situation

Latitude 48.1646 °(N)
Longitude 11.7397 °(E)
Altitude 523 m
Time zone UTC+1

Project settings

Albedo 0.20

Weather data

LessonNo 1

Meteonorm 9.0 dll, Sat=42% - Synthetic

System summary

Grid-Connected System

Simulation for year no 10

Orientation #1

Fixed plane

Tilt/Azimuth 15 / 0 °

Sheds on ground

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules 1272 units
Pnom total 687 kWp

Inverters

Nb. of units 2 units
Total power 704 kWac
Pnom ratio 0.98

Results summary

Simulation step Hourly

Produced Energy 690293 kWh/year Specific production 1005 kWh/kWp/year Perf. Ratio PR 74.31 %

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General parameters

Grid-Connected System

Sheds on ground

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 15 / 0 °

Sheds configuration

Nb. of sheds 64 units

Identical arrays

Shading limit angle

Limit profile angle 64.8 °

Sizes

Sheds spacing 5.00 m

Sensitive width 4.58 m

GCR Shading 91.9 %

Top inactive band 0.02 m

Bottom inactive band 0.02 m

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic

Model CP9612MDGDC-540

(Original PVsyst database)

Unit Nom. Power 540 Wp
Number of PV modules 1272 units
Nominal (STC) 687 kWp
Modules 53 units x 24 In series

At operating cond. (50°C)

Pmpp 629 kWp
U mpp 904 V
I mpp 696 A

Total PV power

Nominal (STC) 687 kWp
Total 1272 modules
Module area 3286 m²
Cell area 3034 m²

Inverter

Manufacturer Generic

Model SG350-HX-20

(Original PVsyst database)

Unit Nom. Power 352 kWac
Number of inverters 2 units
Total power 704 kWac
Operating voltage 500-1450 V
Pnom ratio (DC:AC) 0.98
Power sharing within this inverter

Total inverter power

Total power 704 kWac
Number of inverters 2 units
Pnom ratio 0.98

Array losses

Array Soiling Losses

Loss Fraction 2.0 %

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 21 mΩ
Loss Fraction 1.50 % at STC

LID - Light Induced Degradation

Loss Fraction 2.00 %

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

Strings Mismatch loss

Loss Fraction 0.10 %

Module average degradation

Year no 10
Loss factor 0.4 %/year
Imp / Vmp contributions 80% / 20%

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year



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Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

System losses

Auxiliary losses

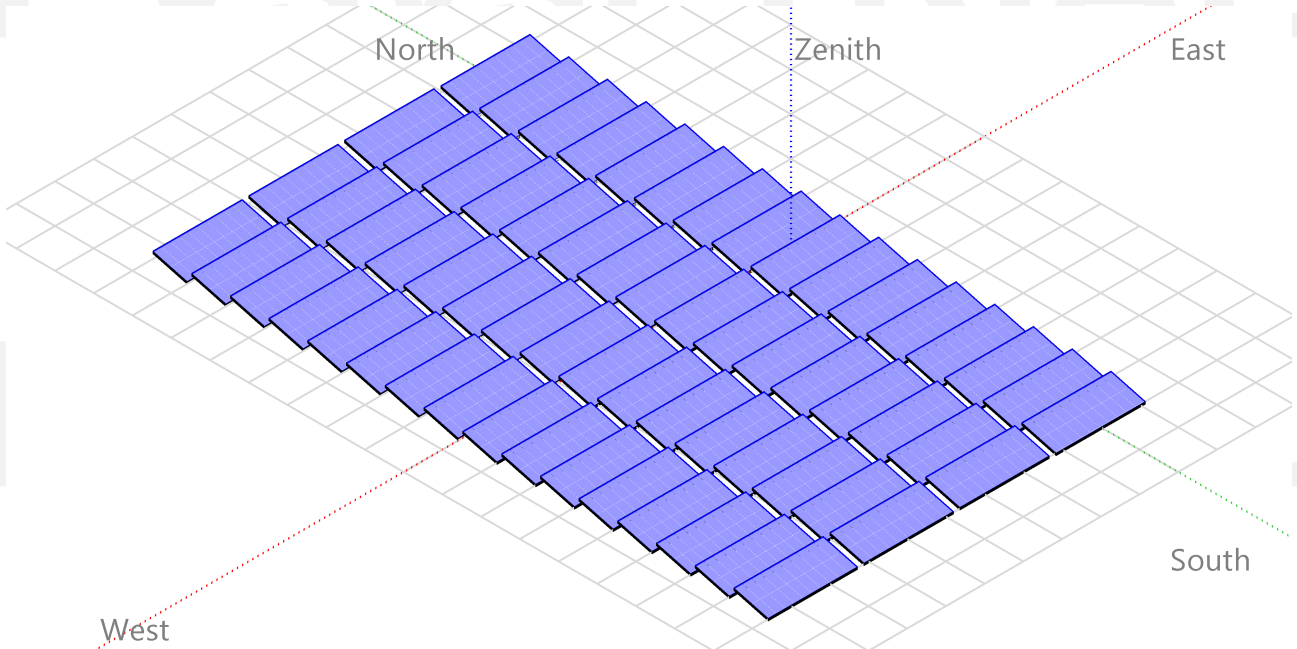


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Near shadings parameter

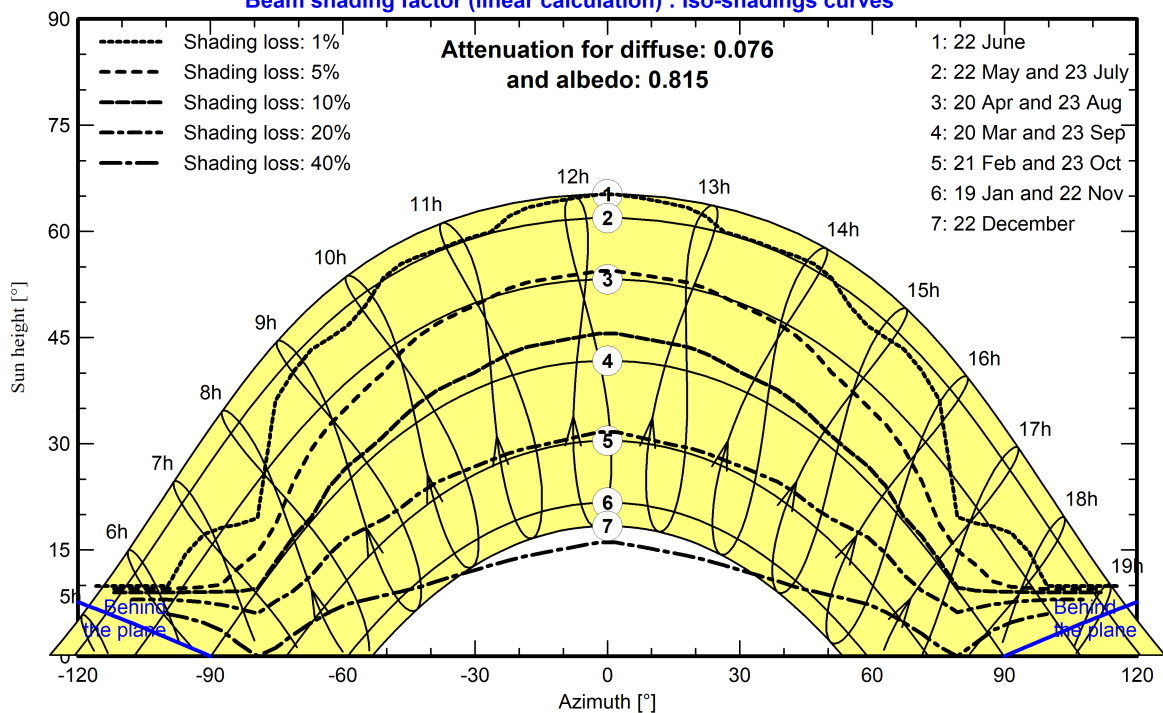
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 15°/ 0°

Beam shading factor (linear calculation) : Iso-shadings curves





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Main results

Simulation step

Hourly

System Production

Produced Energy

690293 kWh/year

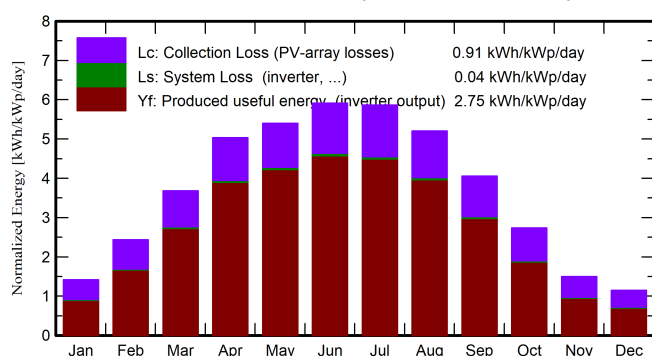
Specific production

1005 kWh/kWp/year

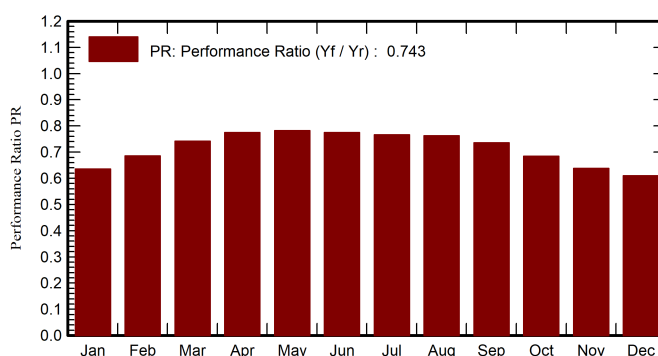
Perf. Ratio PR

74.31 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	31.7	18.79	-0.11	43.9	31.0	19705	19182	0.636
February	51.5	24.60	0.82	68.1	51.5	32668	32064	0.685
March	95.5	43.62	4.80	114.1	95.1	58983	58129	0.742
April	136.6	55.28	9.70	151.0	135.6	81433	80345	0.774
May	161.3	80.33	13.72	167.4	154.0	91202	89964	0.782
June	174.5	80.00	17.51	177.4	164.8	95654	94362	0.774
July	176.9	73.59	19.00	181.8	169.0	96995	95686	0.766
August	151.3	76.25	18.51	161.3	146.6	85644	84473	0.763
September	105.5	46.01	13.90	121.8	105.0	62418	61523	0.736
October	66.5	32.29	9.41	84.8	66.3	40584	39897	0.685
November	33.1	17.63	4.40	44.9	32.5	20244	19699	0.638
December	25.2	15.79	0.99	35.7	24.5	15409	14968	0.610
Year	1209.6	564.17	9.44	1352.3	1176.1	700939	690293	0.743

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

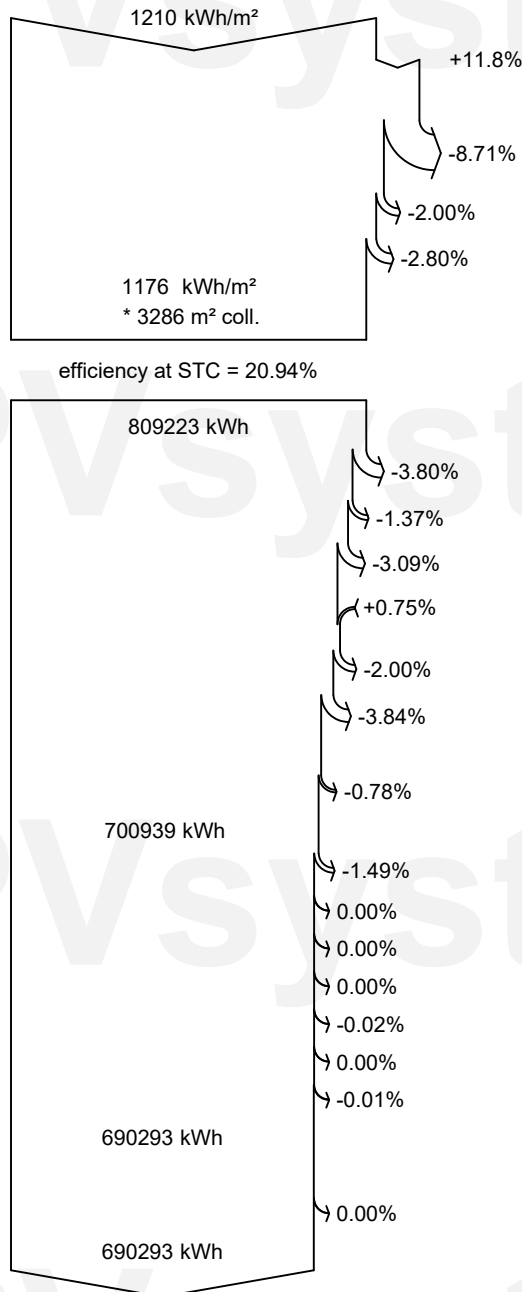
PR Performance Ratio



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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Near Shadings: irradiance loss

Soiling loss factor

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

Module Degradation Loss (for year #10)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

LID - Light induced degradation

Mismatch loss, modules and strings
(including 1.7% for degradation dispersion)

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Auxiliaries (fans, other)

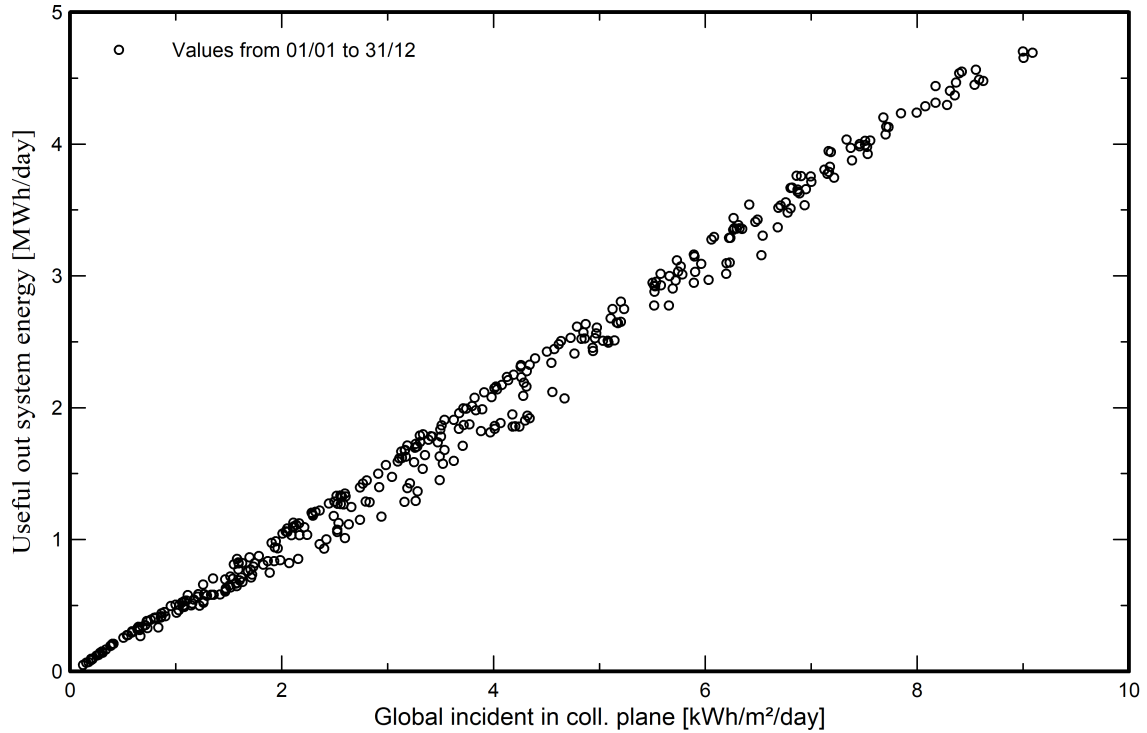
Energy injected into grid



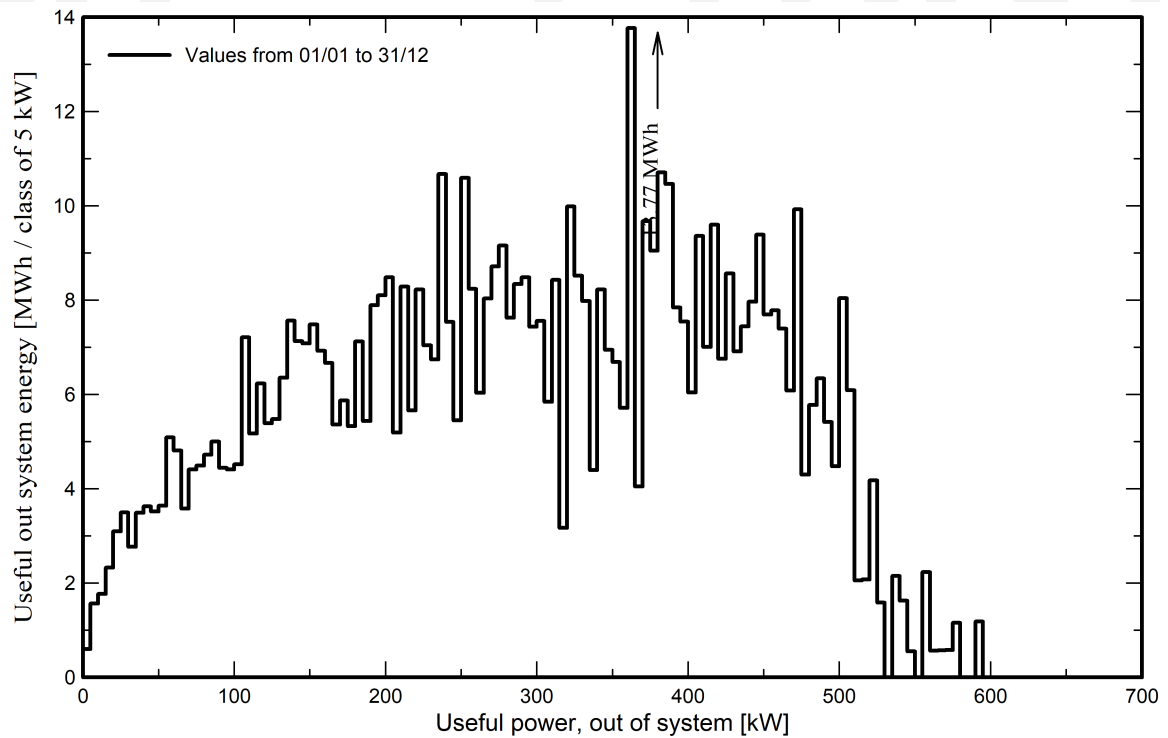
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

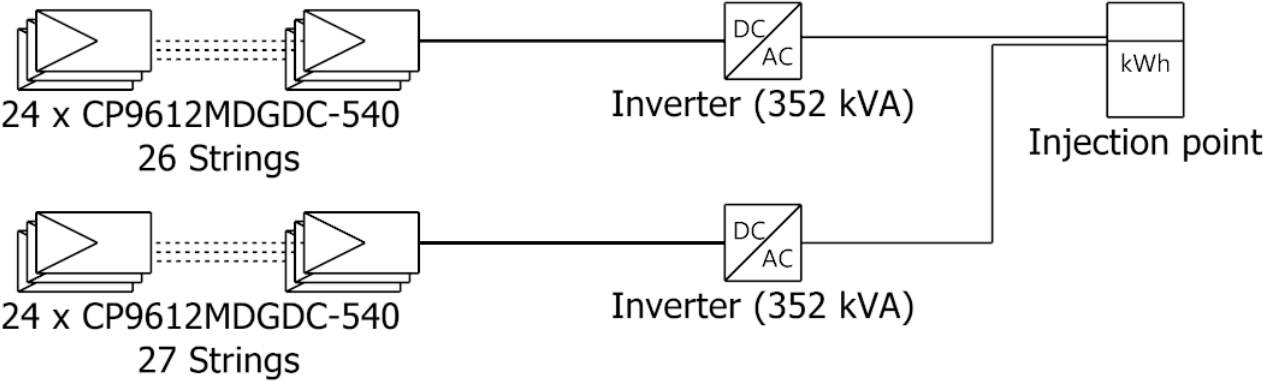




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Single-line diagram



PV module	CP9612MDGDC-540
Inverter	SG350-HX-20
String	24 x CP9612MDGDC-540

DHL

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